MEC V1.0 - Modular Embedded Controller

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Uniformar librería Schematics

Uniformar librería Footprints según IPC

Legend

General comment such function title, configurations, datasheets, ...

Text to be added to silkscreen

Design references

Warning text

DNP Do-not-place

Notes to generate the board layout

U_Block diagram
Block diagram.SchDoc

Main
Main.SchDoc

Title:	Project overview.SchDoc		
Project:	MEC.PrjPcb		
Revision:	1.0	Sheet:	1 of 10
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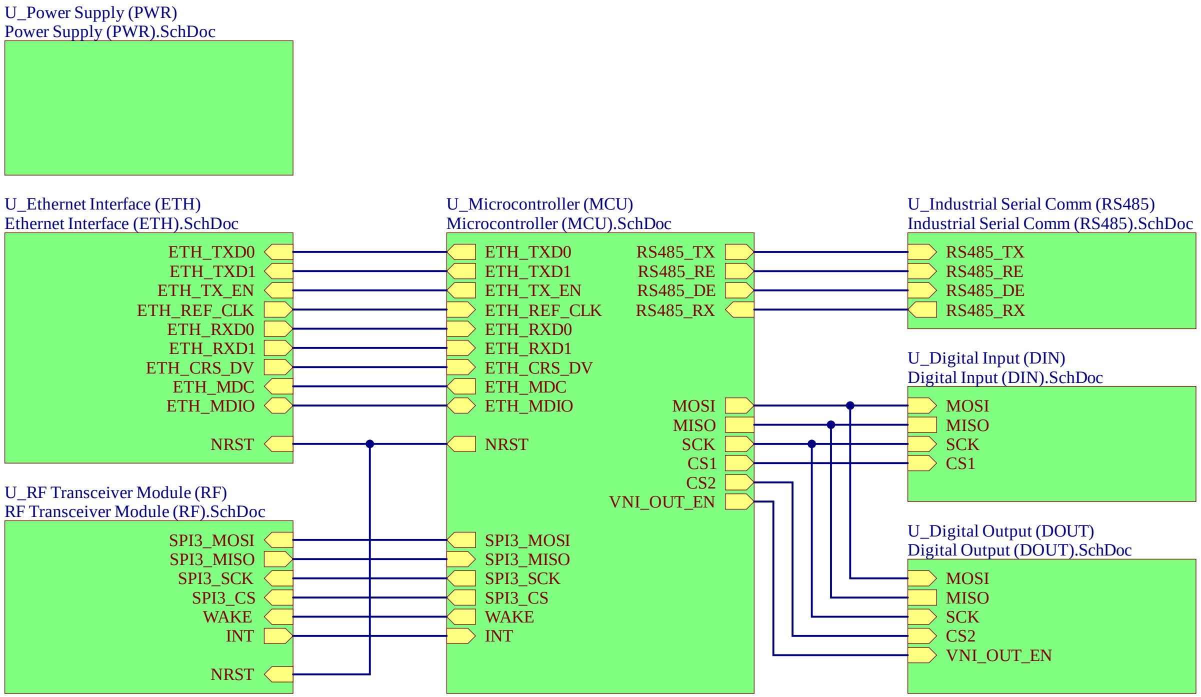
Draw by:	Ing. Lucas Kirschner		

Reservado diagrama de bloques

Title:	Block diagram.SchDoc
Project:	MEC.PrjPcb
Version 1.0	Sheet: 2 of 10
Size: A4	Date: 12/11/2025
Draw by: Ing. Lucas Kirschner	



Title:	Main.SchDoc
Project:	MEC.PrjPcb
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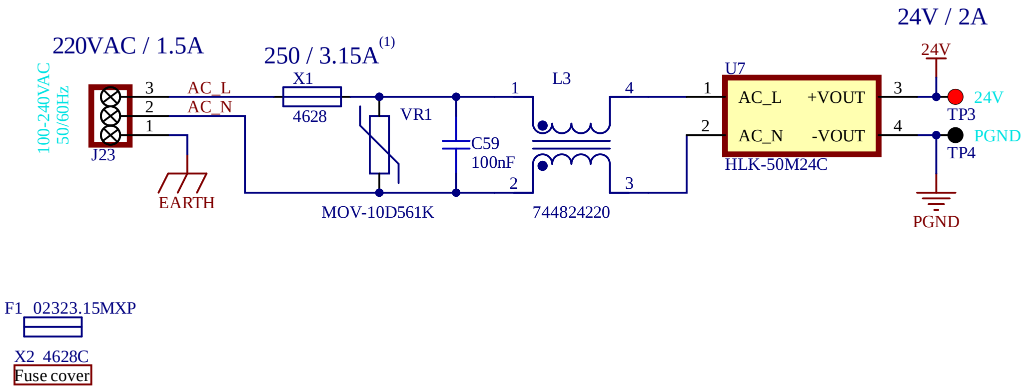
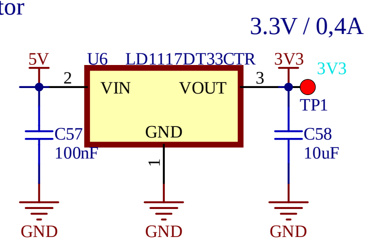
Draw by: Ing. Lucas Kirschner	

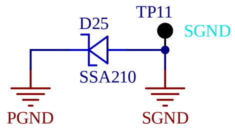
220VAC to 24VDC AC/DC converter



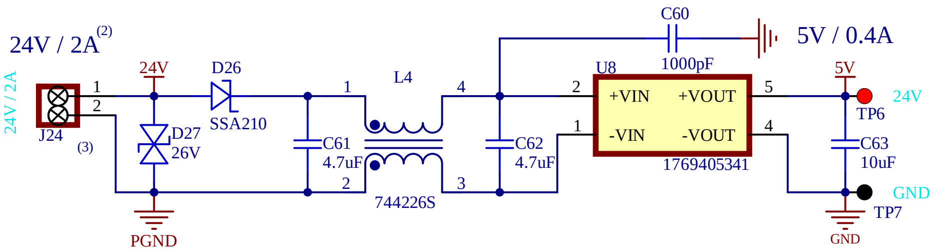
3V3 linear regulator



GND NET Tie



24V to 5V DC/DC isolated converter



References

1) Fusible de acción media para la entrada de red (AC). Modelo recomendado: Littelfuse 02323.15MXP.

2) Corriente máxima total del riel de 24 V: 2 A. Cuando el conector de 24 V actúa como salida (módulo HLK-50M24C instalado), este valor corresponde a la suma de todos los consumos del sistema. La corriente disponible en el conector disminuirá si las salidas digitales entregan potencia.

3) El conector de 24 V puede funcionar como entrada o salida, según la presencia o ausencia del módulo HLK-50M24C en el circuito. No aplicar 24 V externos si el módulo HLK-50M24C está instalado.

Title: Power Supply (PWR).SchDoc	
Project: MEC.PrjPcb	
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Draw by: Ing. Lucas Kirschner	

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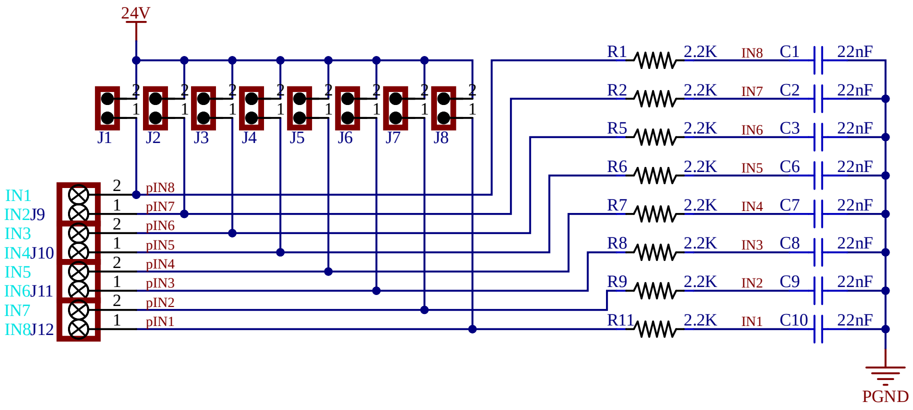
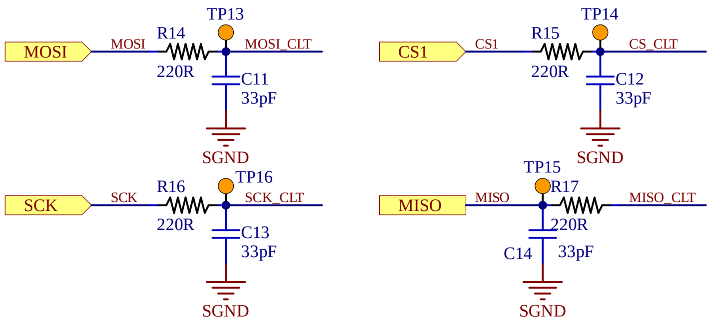
Digital input connector



SPI signals

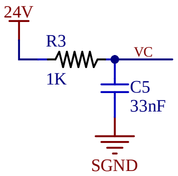


References

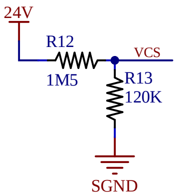
See https://www.st.com/en/protections-and-emi-filters/sclt3-8bt8
See https://www.st.com/en/evaluation-tools/x-nucleo-plc01a1
See https://www.st.com/en/evaluation-tools/steval-ifp031v1
See https://www.st.com/en/evaluation-tools/steval-ifp007v1

High speed digital input current limiter

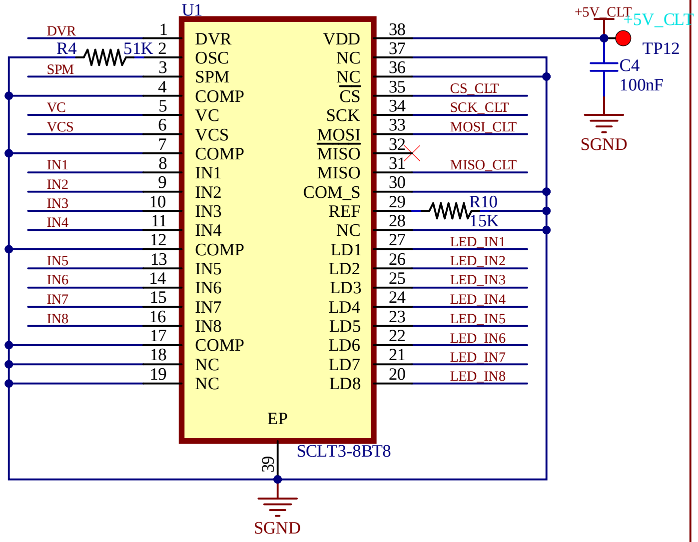
24 V sensor power supply



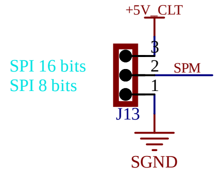
24 V sensor power supply sensing input



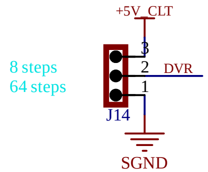
VDD máx power supply: 12mA



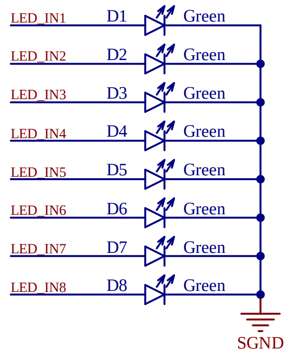
SPI 8/16-bit selection



Divider ratio selector of the digital input filters



LED input driver



Title: Digital Input (DIN).SchDoc	
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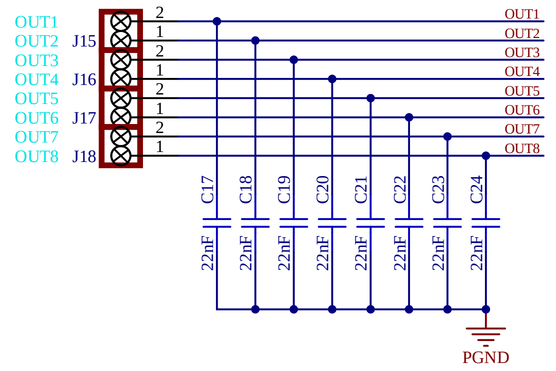
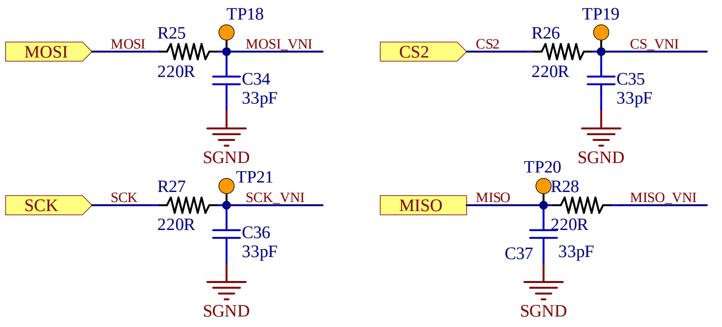
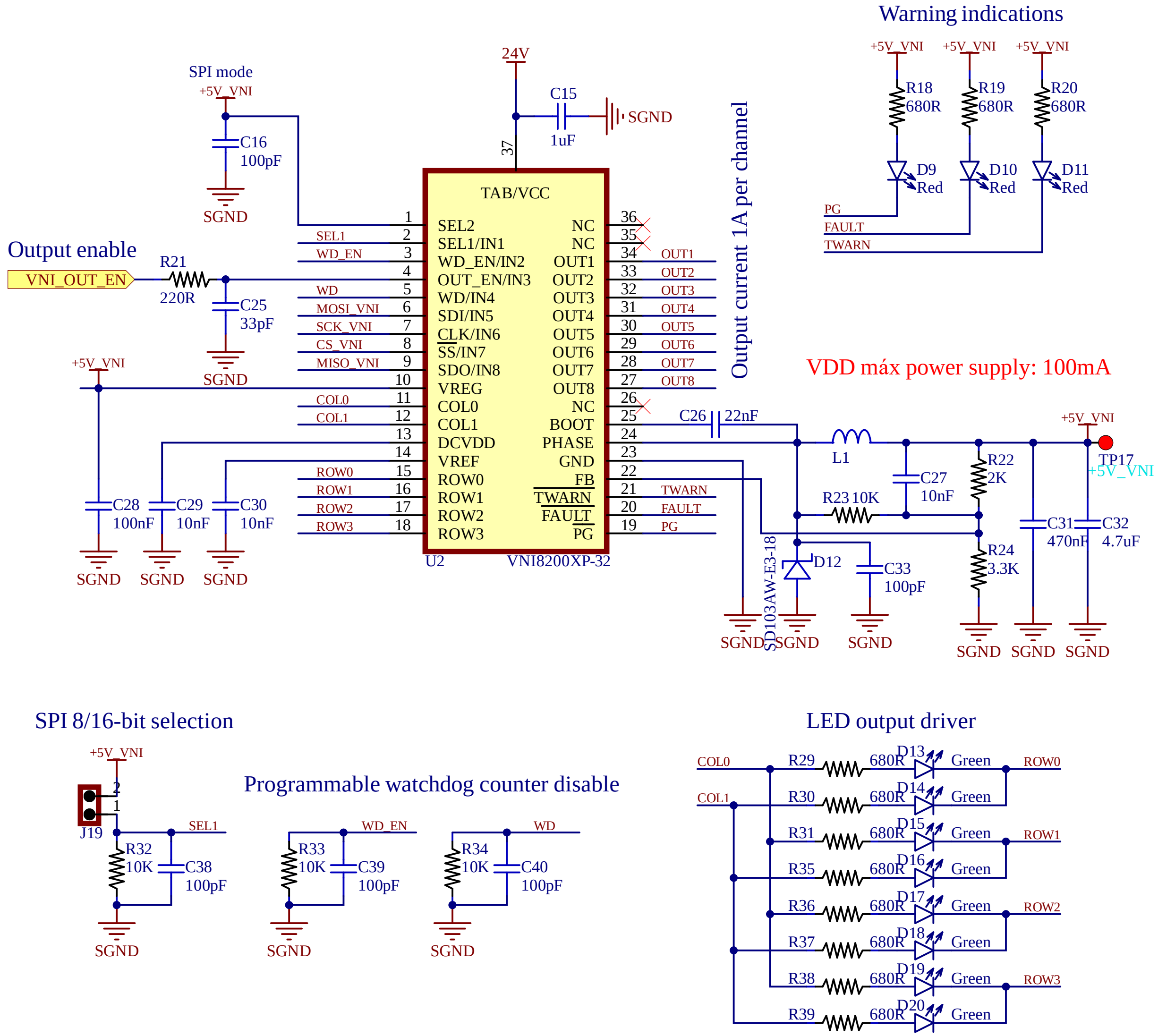
Digital output connector



SPI signals



Octal high-side smart power interface



References

See https://www.st.com/en/power-management/vni8200xp-32
See https://www.st.com/en/evaluation-tools/x-nucleo-plc01a1
See https://www.st.com/en/evaluation-tools/steval-ifp032v1
See https://www.st.com/en/evaluation-tools/steval-ifp022v1

Title:	Digital Output (DOUT).SchDoc
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Revision: 1.0	Sheet: 6 of 10
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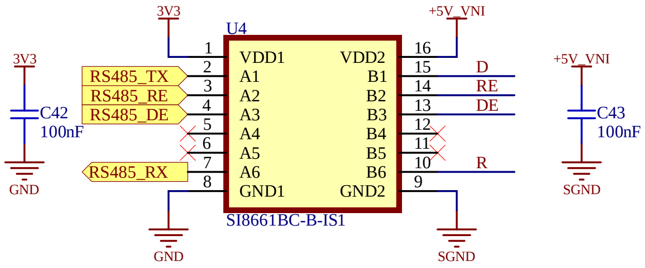
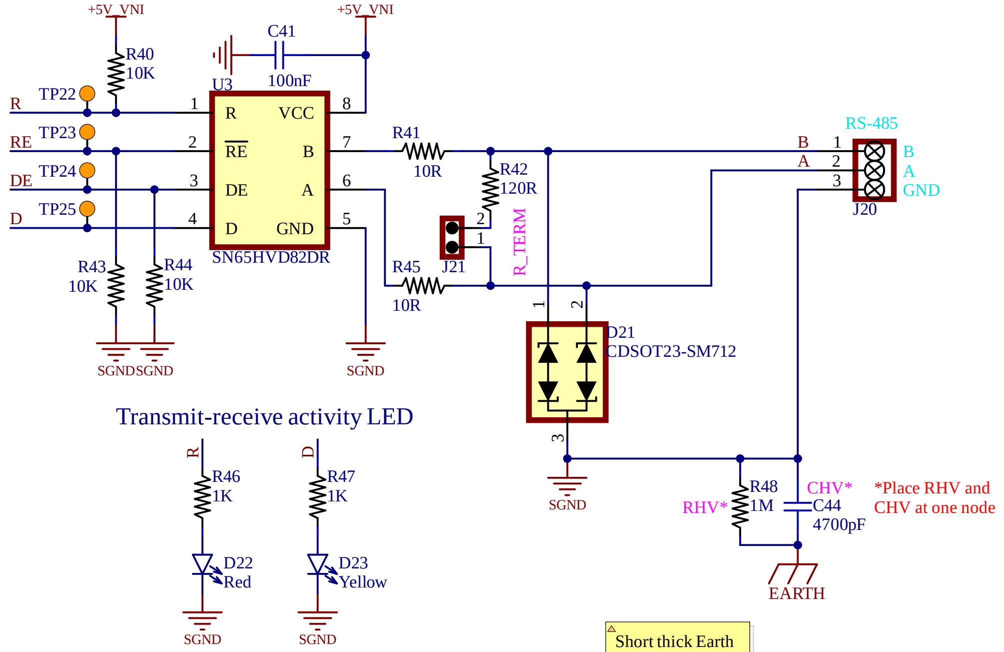
Draw by:	Ing. Lucas Kirschner

Digital isolator



RS485 interface



Short thick Earth wire or Chassis

Protective Earth island

A: Driver output or receiver input (complementary to B)
B: Driver output or receiver input (complementary to A)
D: Driver data input
DE: Driver enable, active high
R: Receive data output
RE: Receiver enable, active low

References

See https://www.ti.com/product/SN65HVD82
See https://www.st.com/en/evaluation-tools/x-nucleo-plc01a1
See https://www.ti.com/tool/RS485-HF-DPLX-EVM
See https://www.ti.com/lit/an/slla070d/slla070d.pdf

Title: Industrial Serial Comm (RS485).SchDoc	
Project: MEC.PrjPcb	
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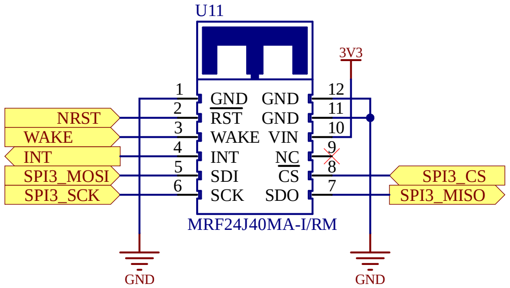
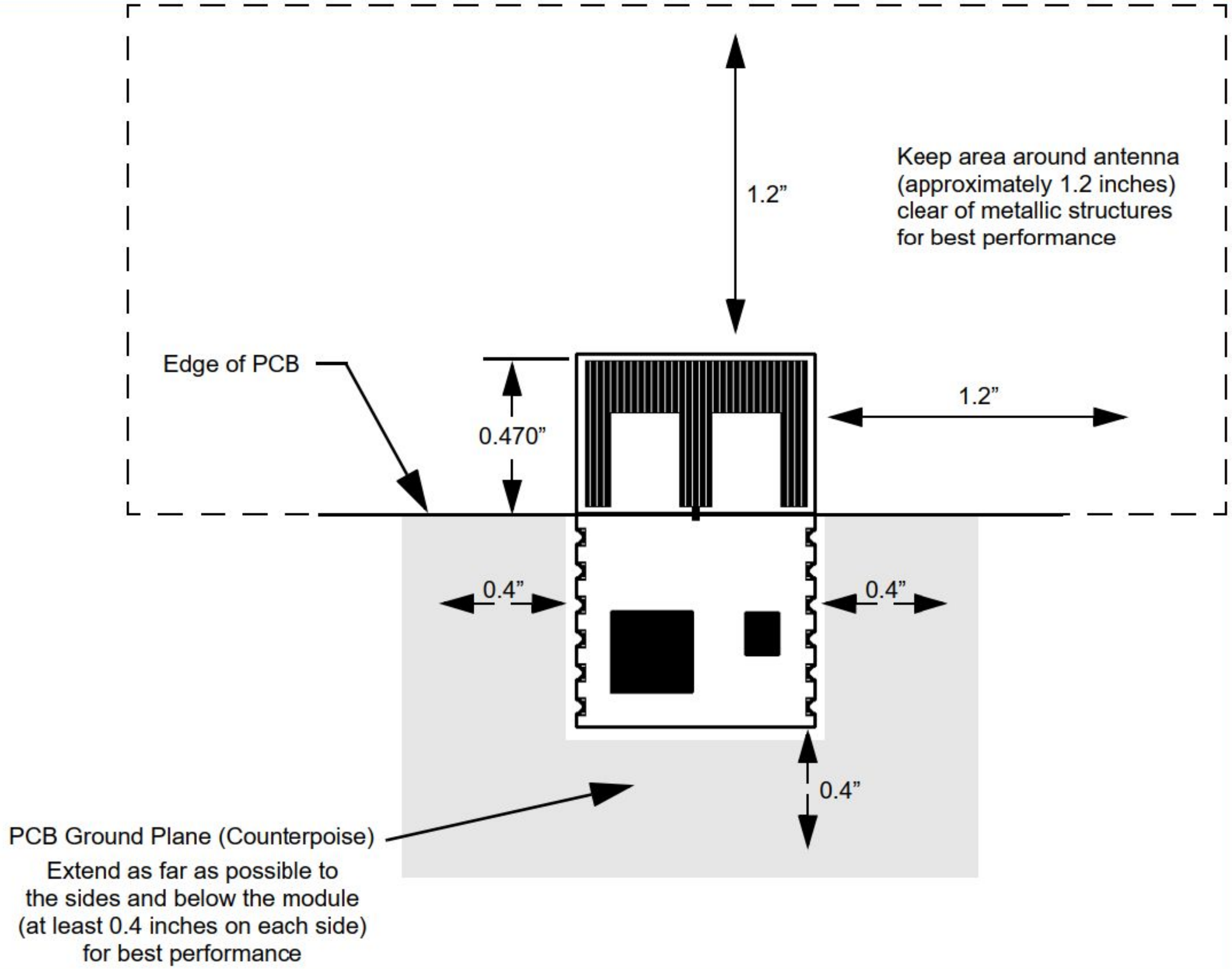
RF Transceiver Module





References

See https://www.microchip.com/en-us/product/MRF24J40MA

Title:	RF Transceiver Module (RF).SchDoc
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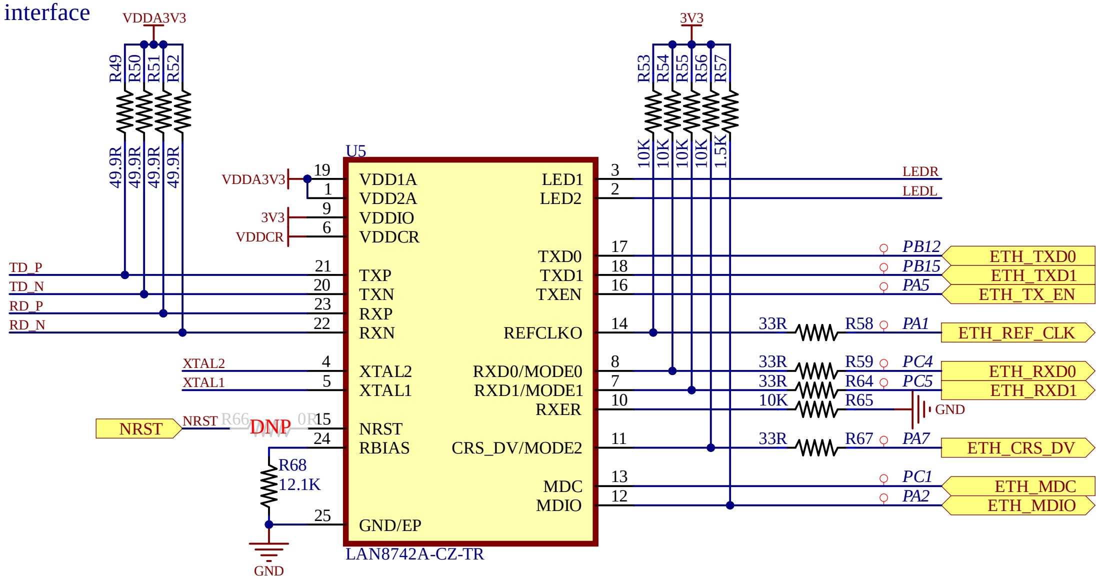
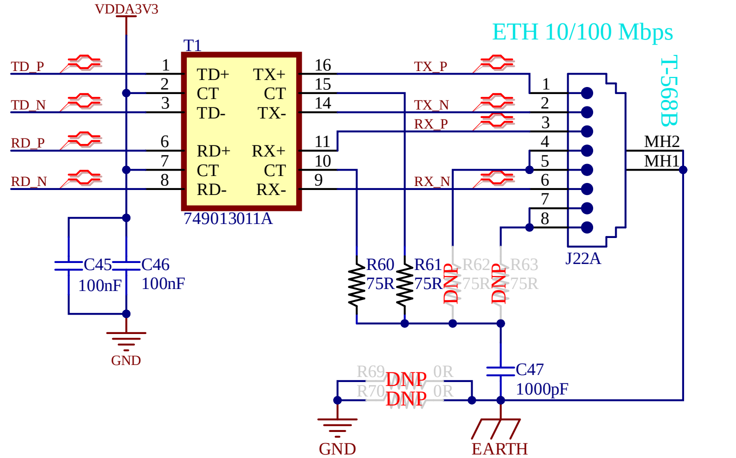
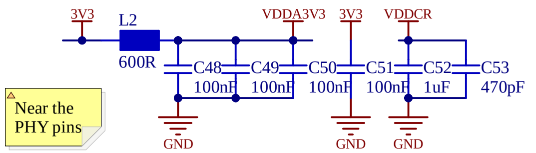
PHY interface



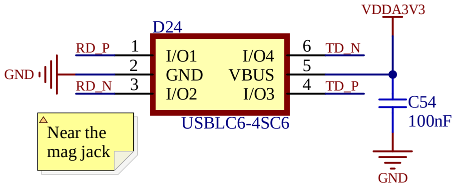
Ethernet transformer & connector



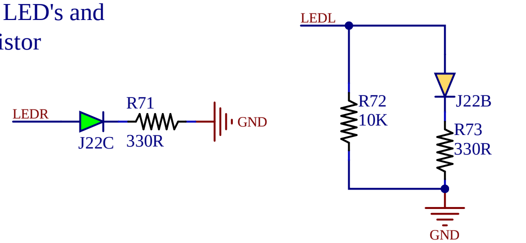
PHY decoupling capacitors



ESD protecction



Activity LED's and strap resistor



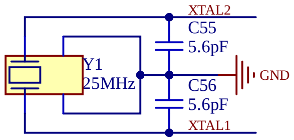
References

- Los resistores de pull up en los pines MODE0, MODE1 y MODE2 son resistores de strap que configuran al LAN8742 en el modo "all capable" con la autonegociación activada.

- LEDL cuenta con un resistor de strap que configura el pin REFCLKO como salida de reloj de referencia para la capa MAC, si se deja flotado o conectado a VDD2A, el pin indica interrupciones nINTSEL.

See https://www.st.com/en/evaluation-tools/nucleo-h563zi
See https://www.microchip.com/downloads/DS_LAN8742_00001989A.pdf

External crystal



Title:	Ethernet Interface (ETH).SchDoc
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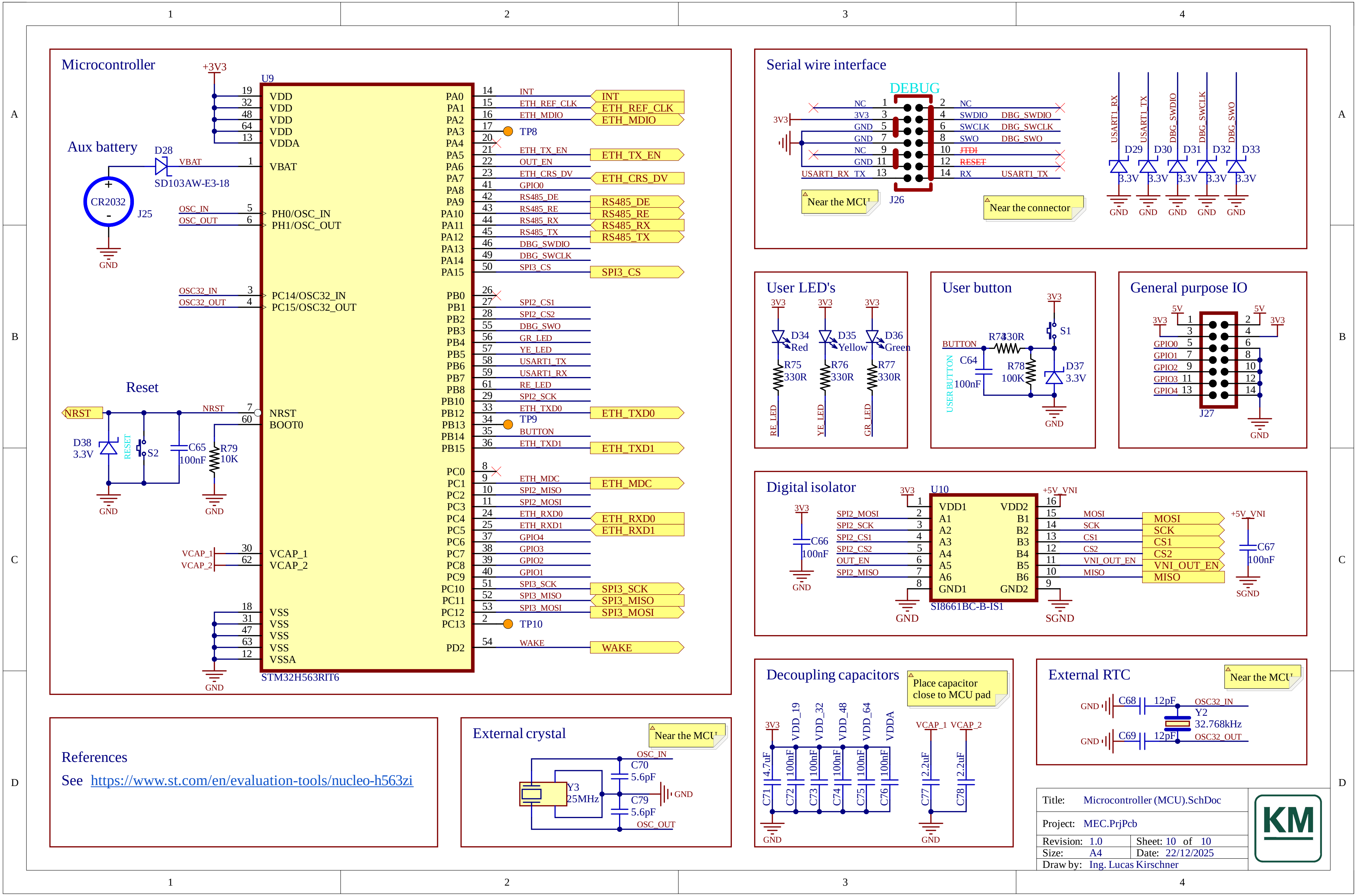

References

See https://www.st.com/en/evaluation-tools/nucleo-h563zi

Title:	Microcontroller (MCU).SchDoc
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Draw by:	Ing. Lucas Kirschner